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(54) **MULTI-PANE PORTABLE GLASS SURROUND FOR PROPANE FIRE PIT**

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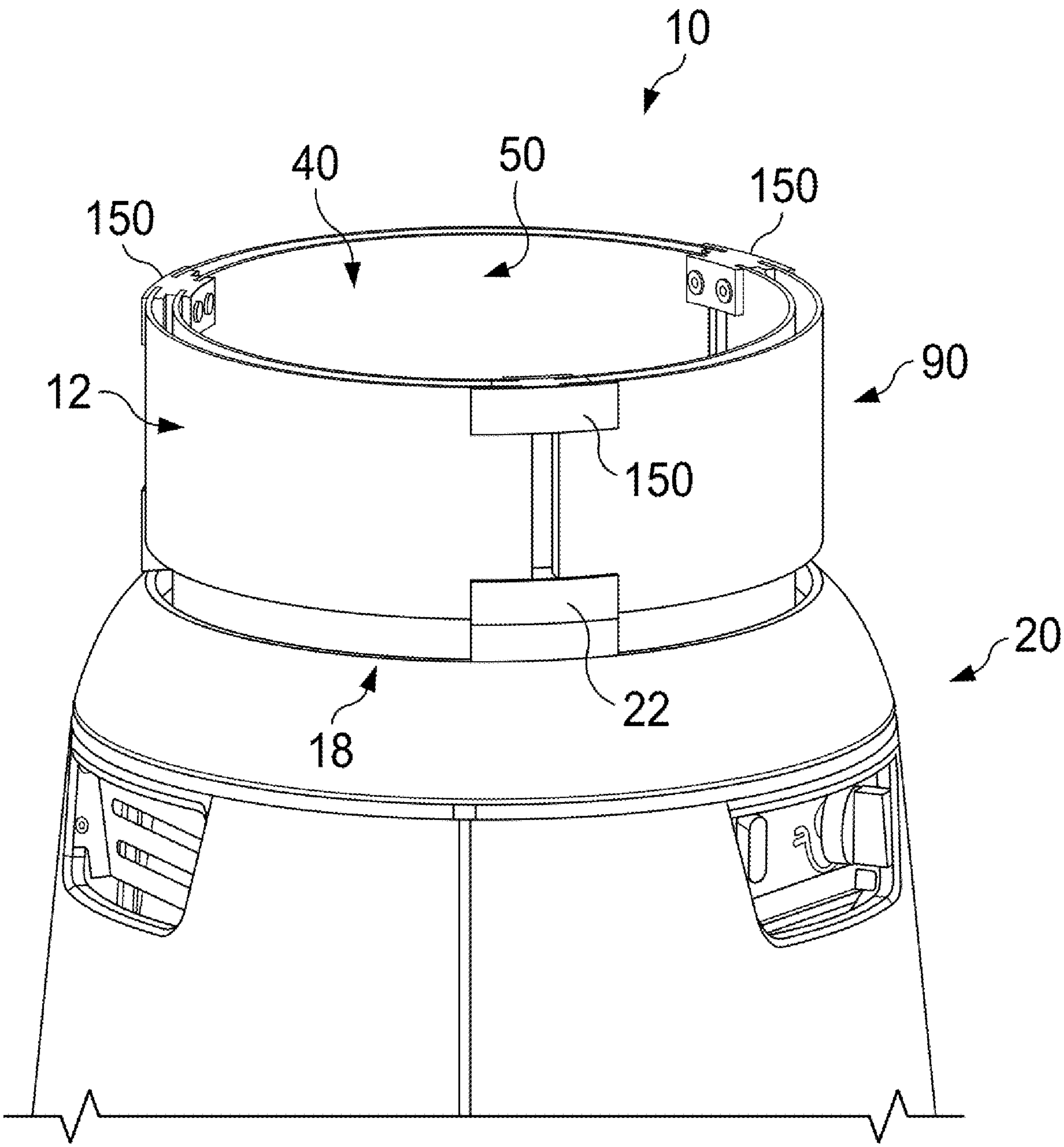
***F24B 3/00*** (2006.01)

(52) **U.S. Cl.**
CPC ...................................... ***F24B 3/00*** (2013.01)

(57) **ABSTRACT**

A fire pit surround having an inner wall and an outer wall surrounding the inner wall.



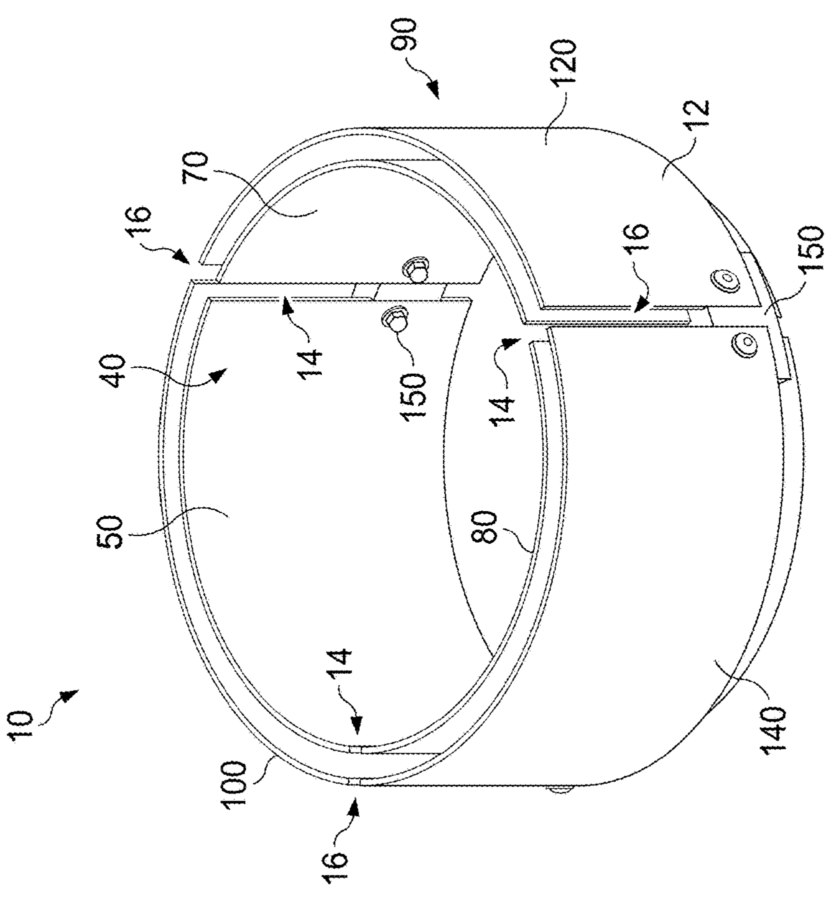

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FIG. 1

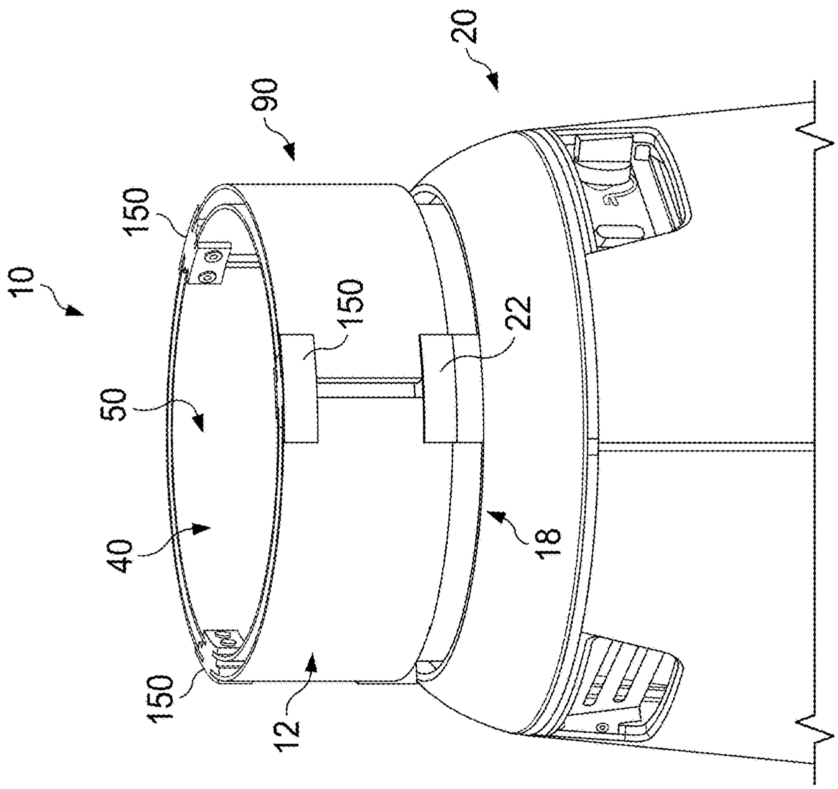
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FIG. 2

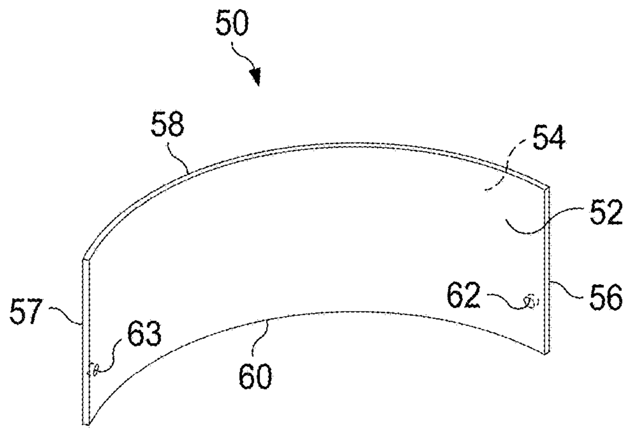



FIG. 3

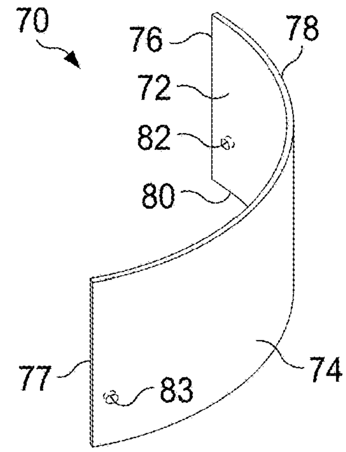



FIG. 5

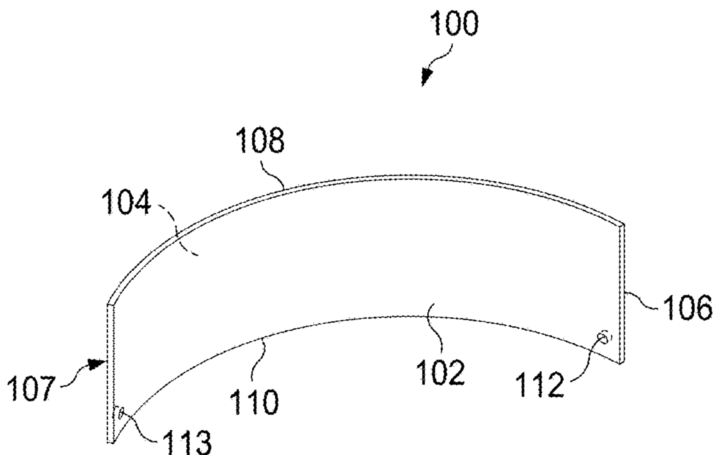
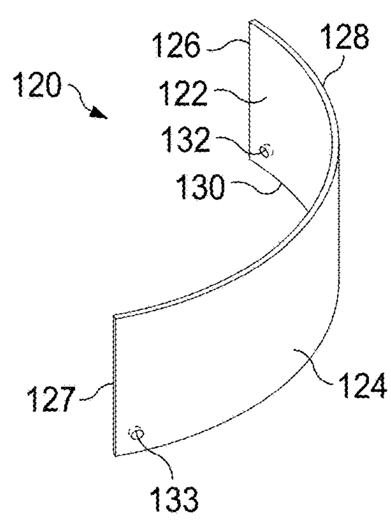



FIG. 4

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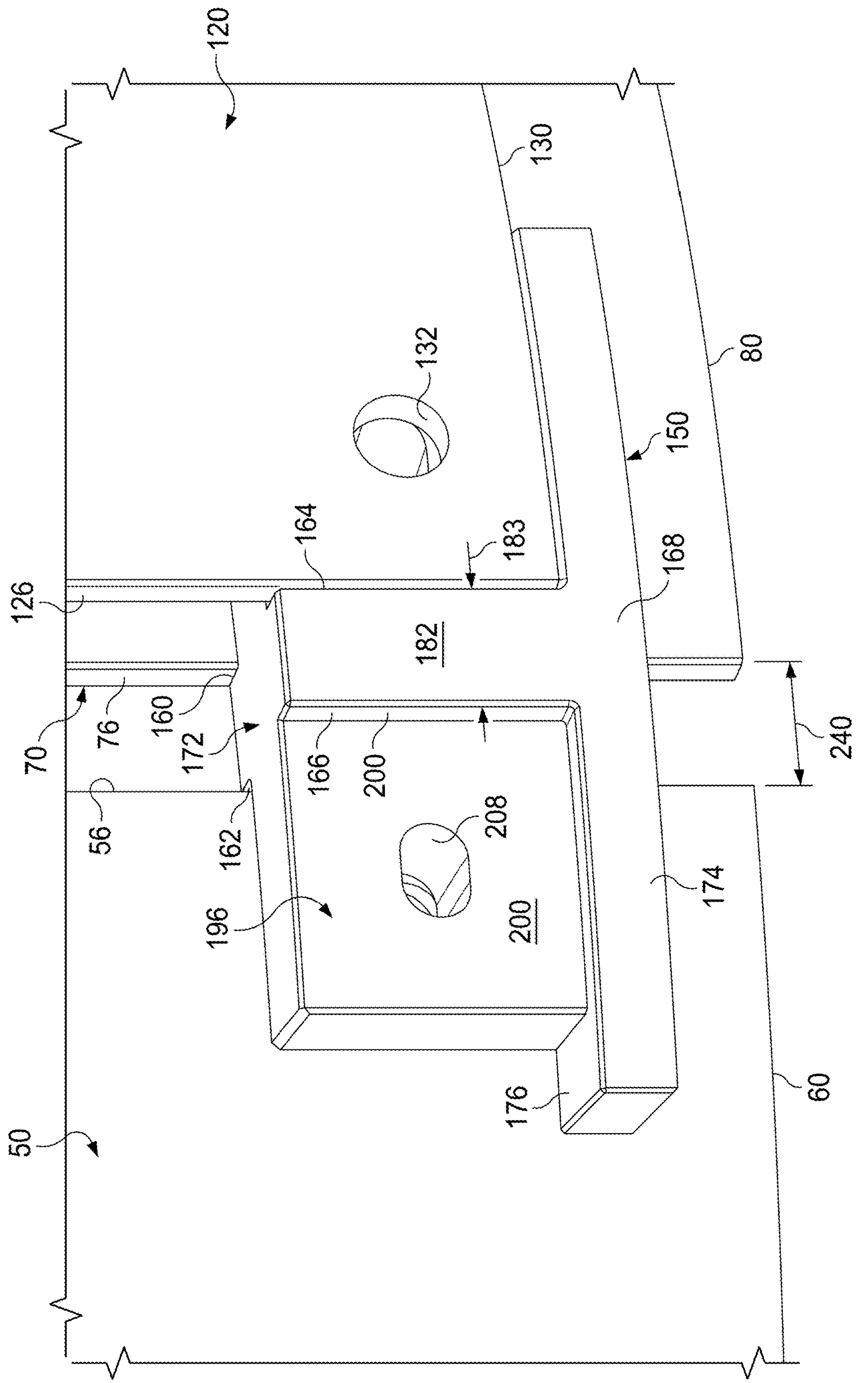
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FIG. 7A

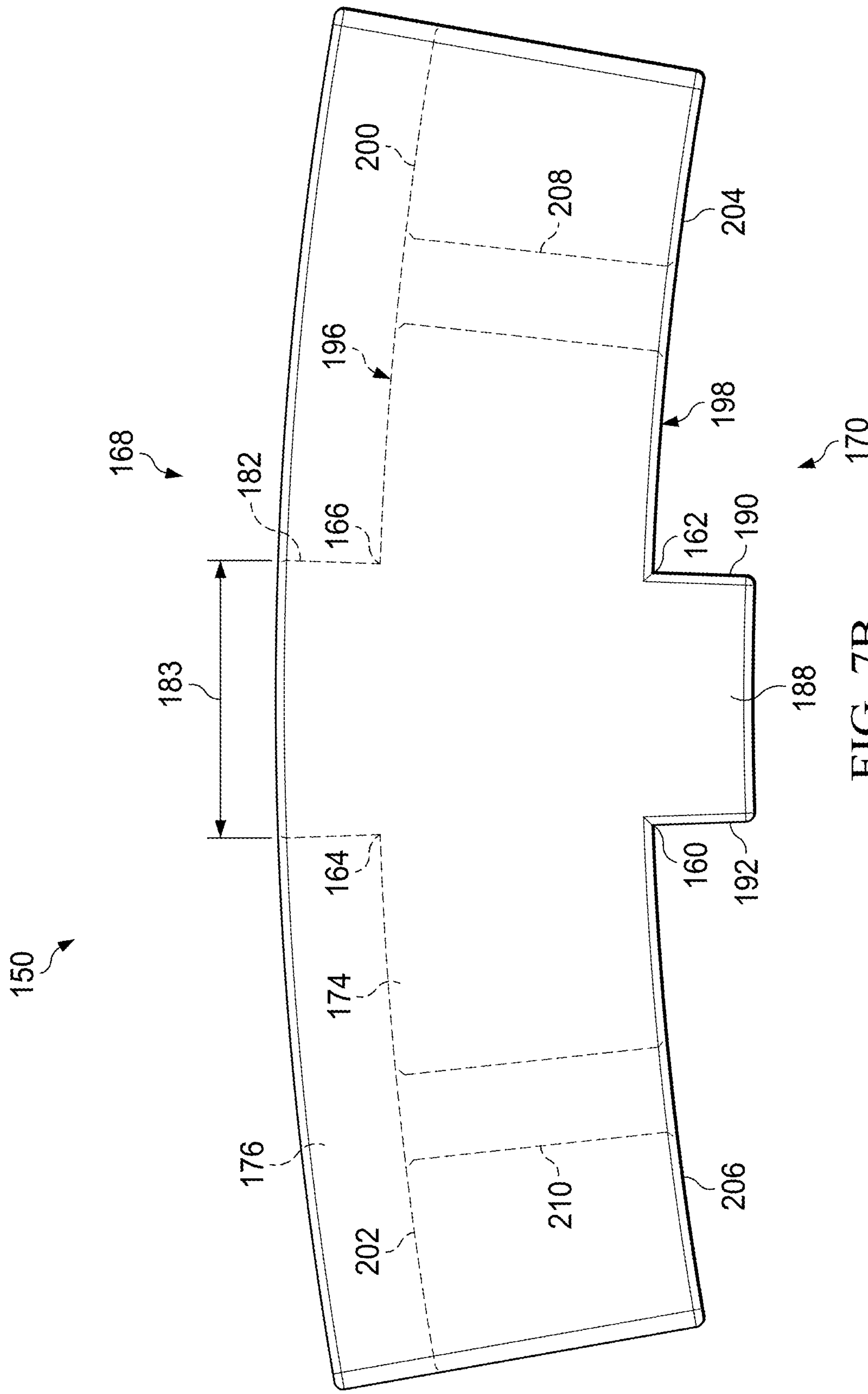
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FIG. 7B

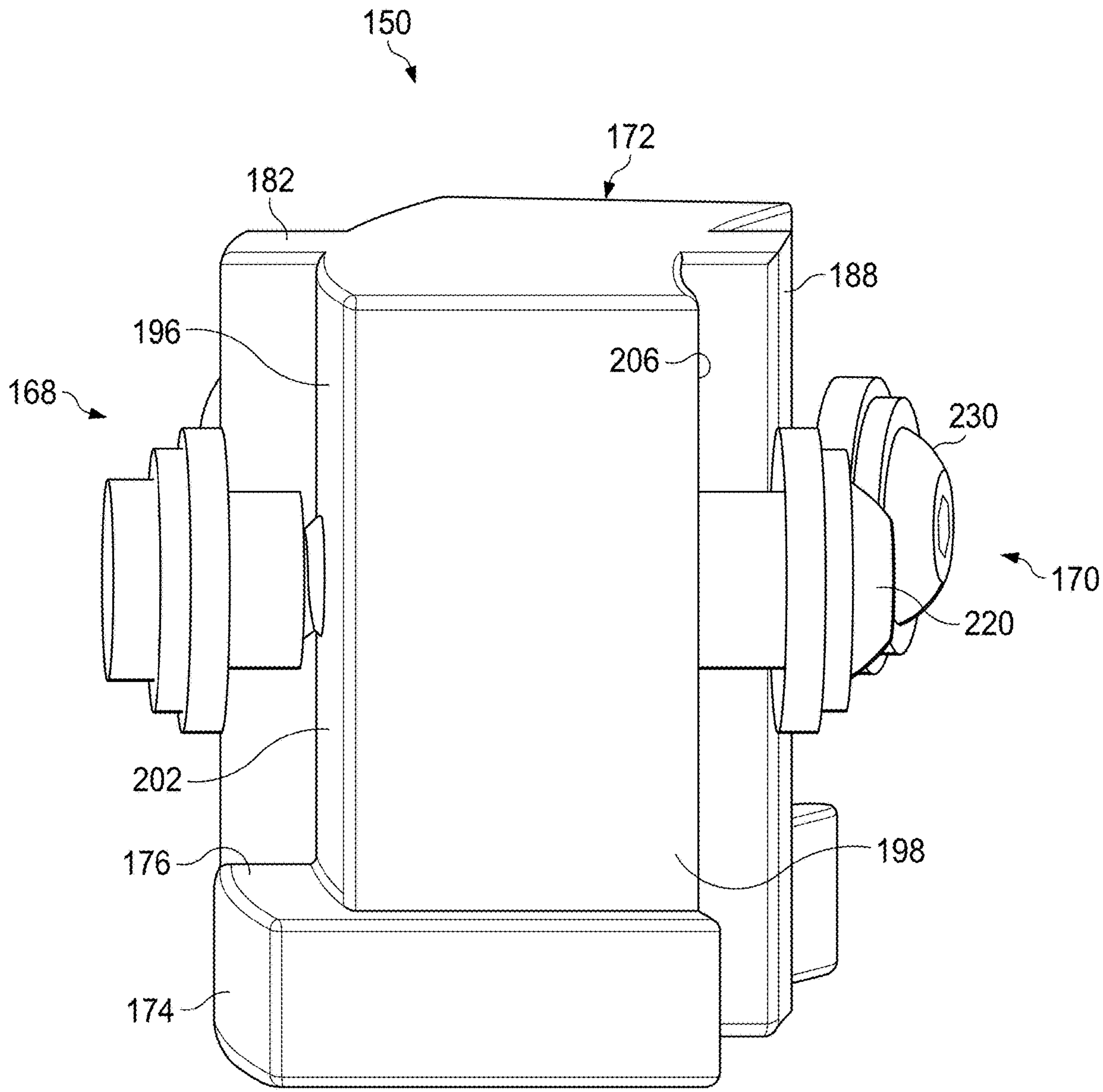


FIG. 8

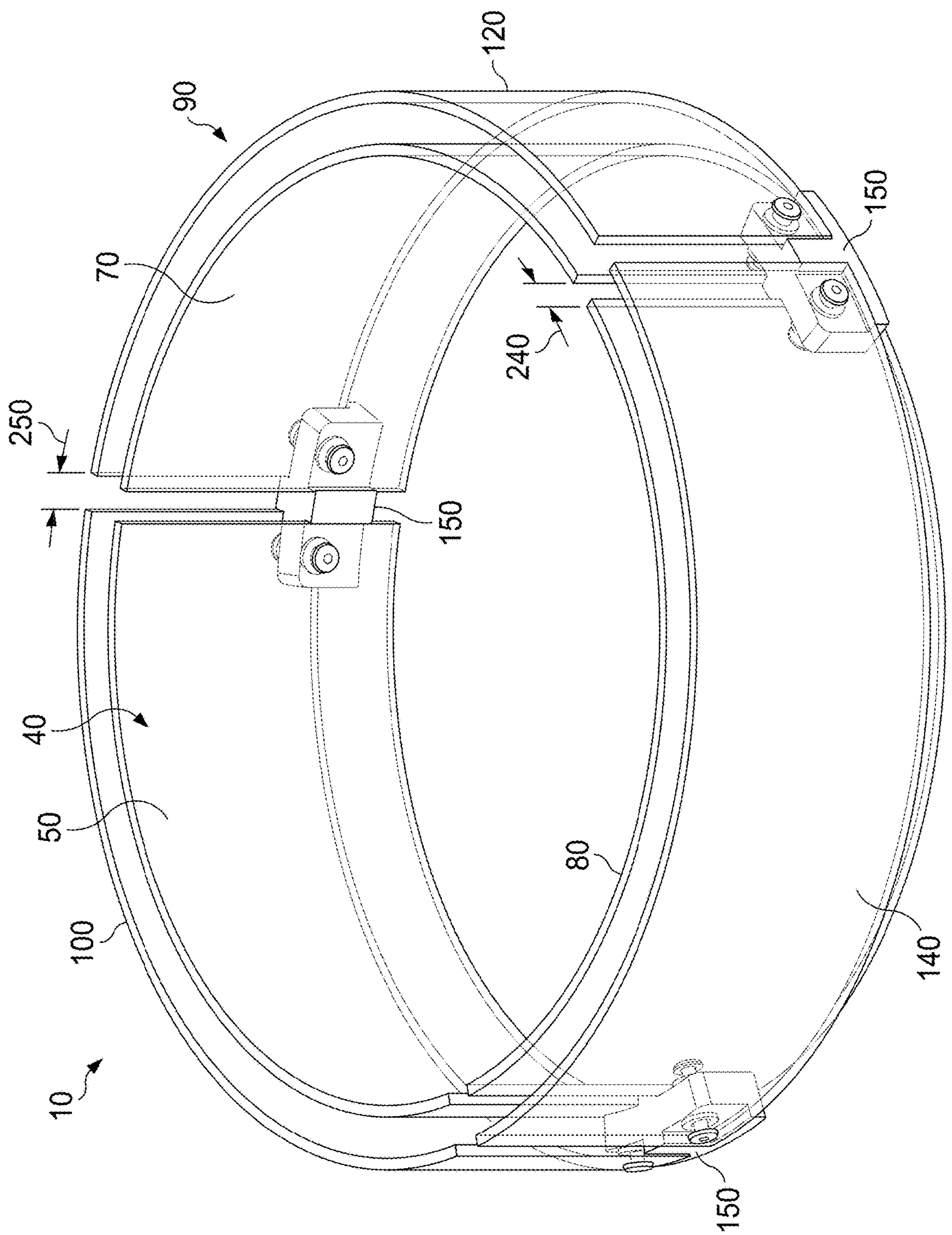


FIG. 9

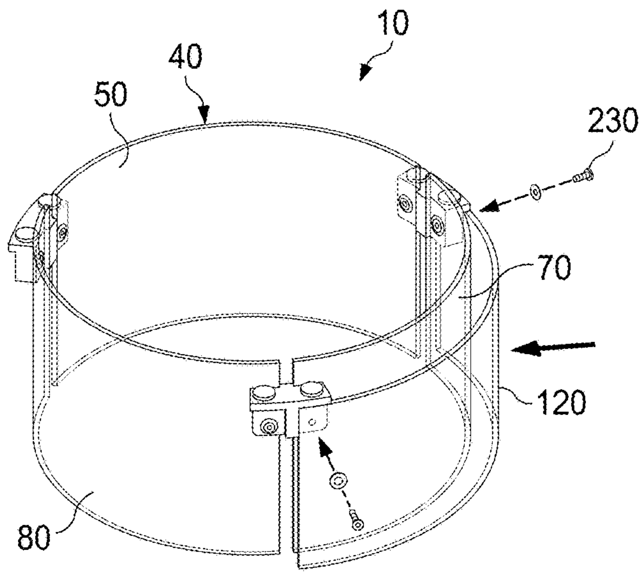


FIG. 10

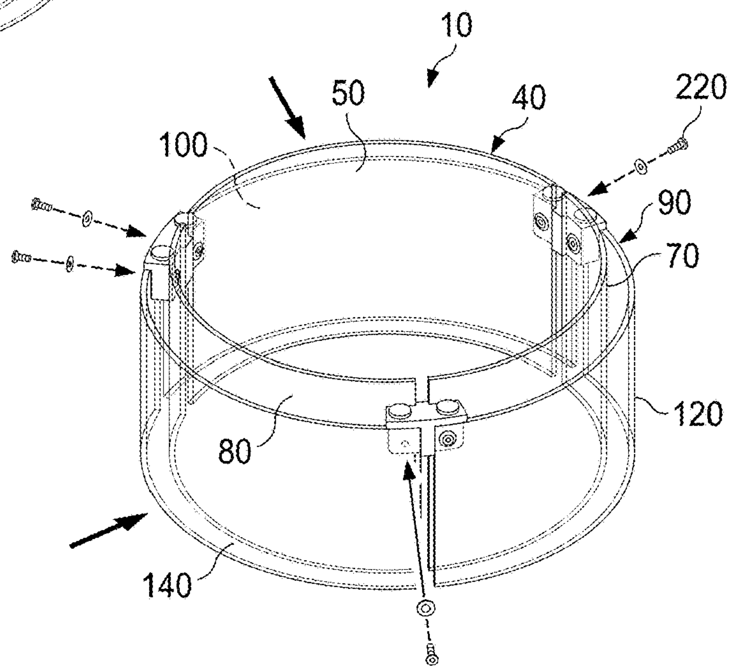


FIG. 11

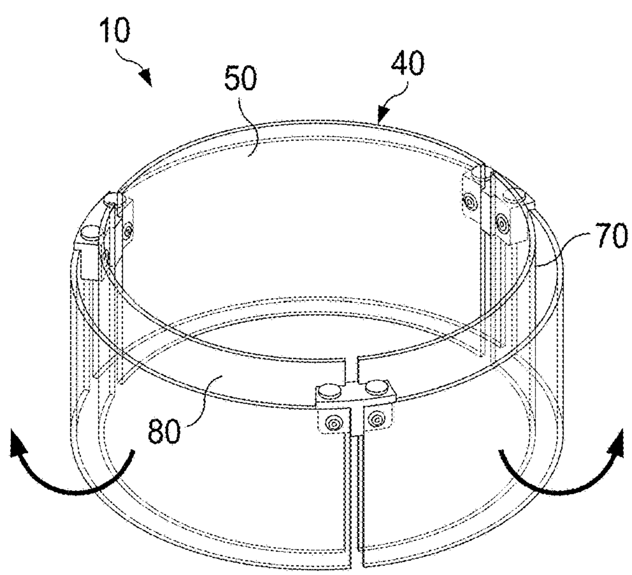


FIG. 12

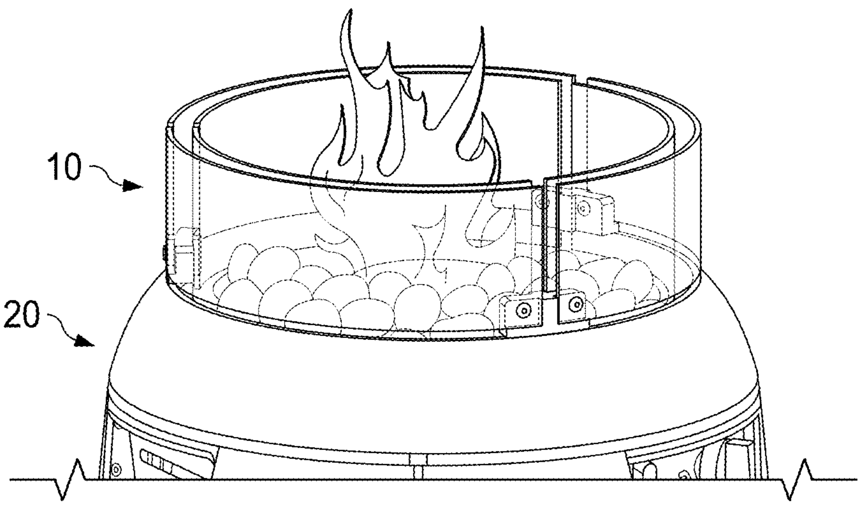


FIG. 13

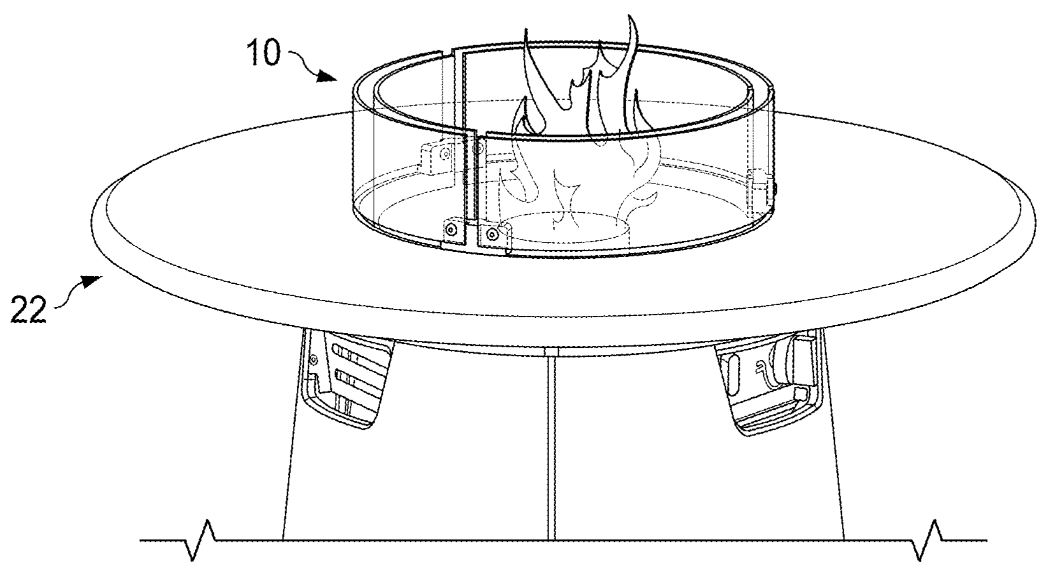


FIG. 14

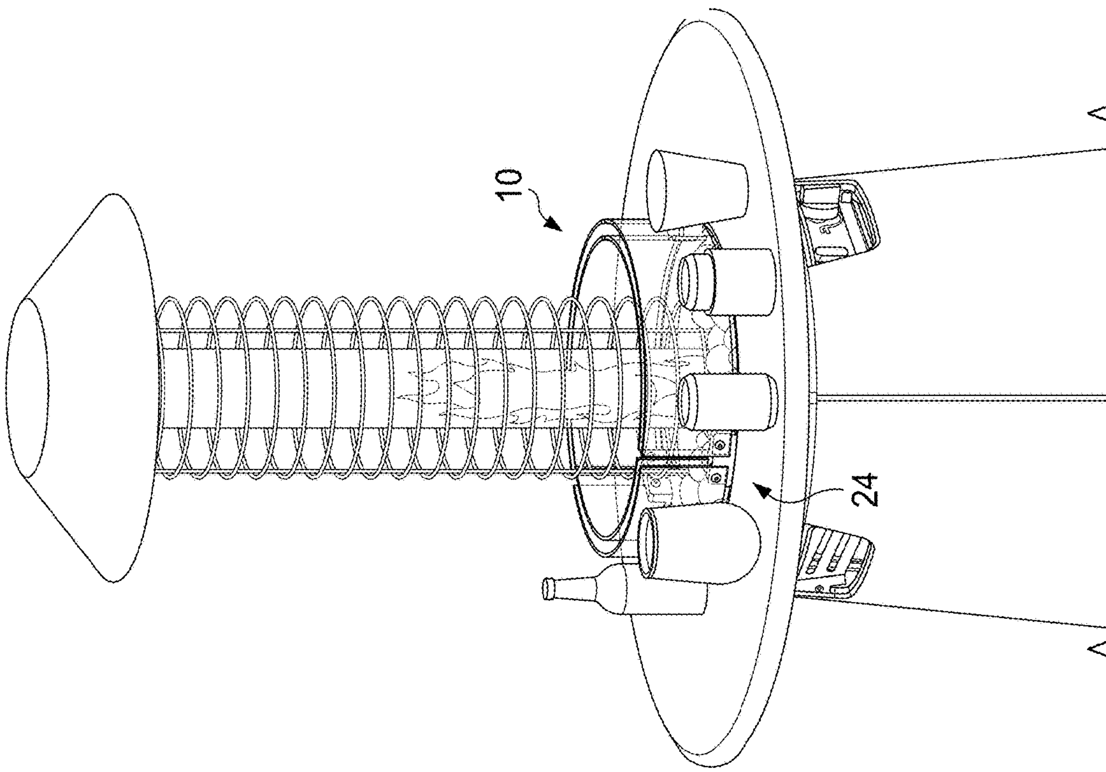


FIG. 16

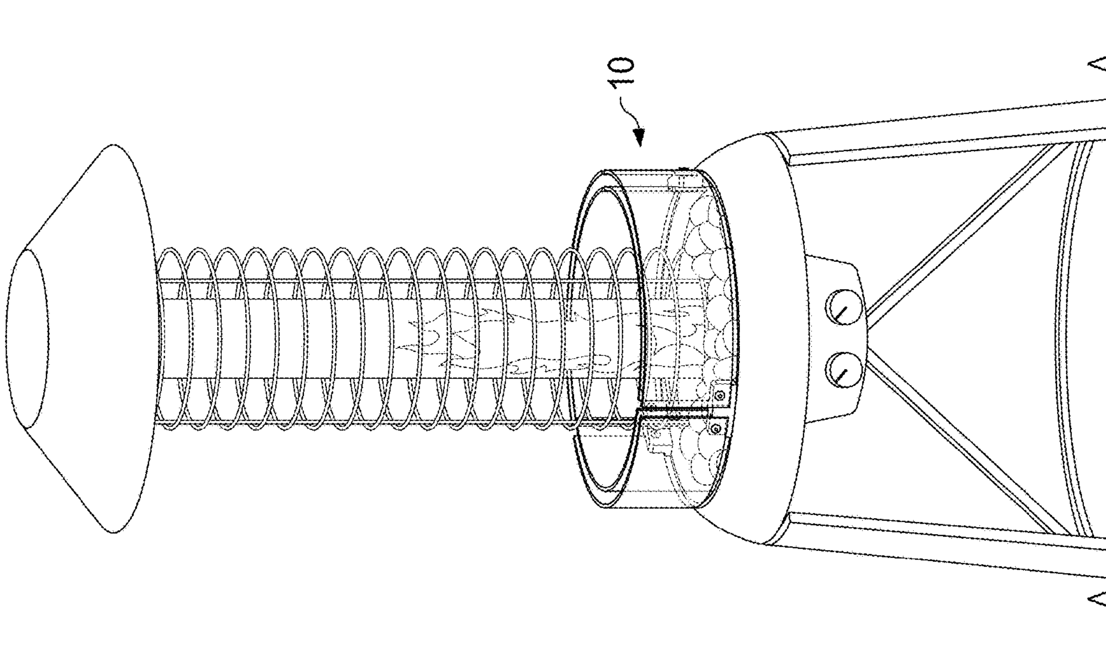


FIG. 15

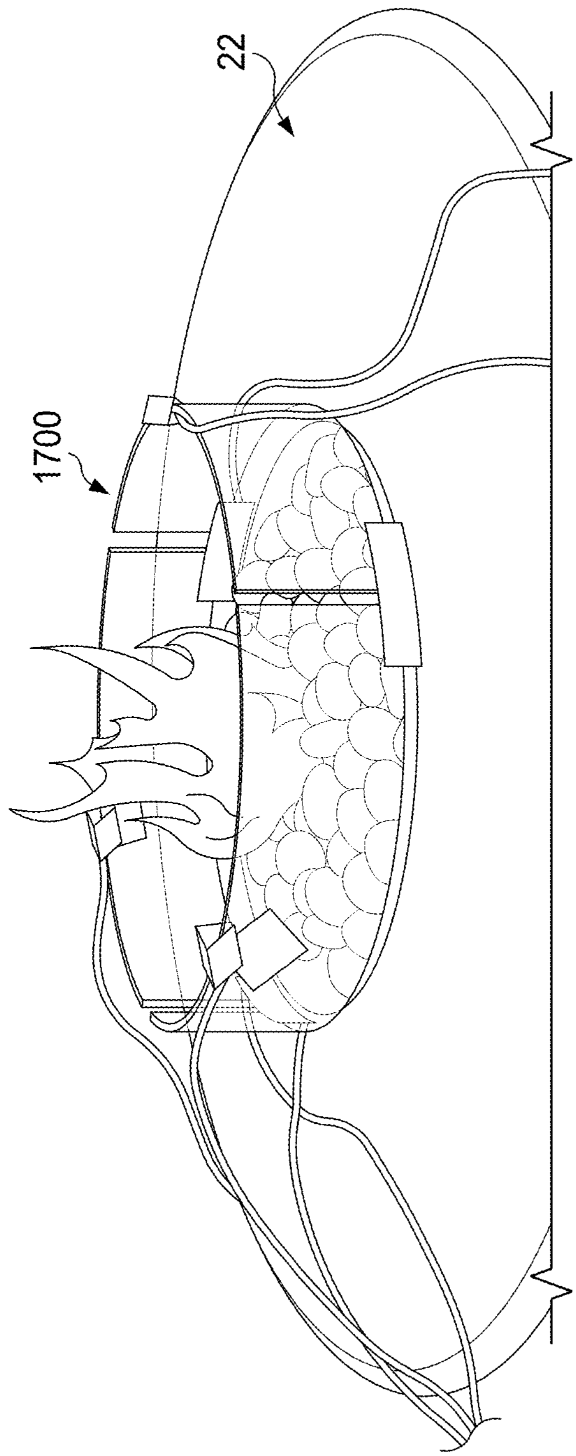


FIG. 17

| TIME | TOP 1 | BOTTOM 1 | TOP 2 | BOTTOM 2 | TOP 3 | BOTTOM 3 |
|---|---|---|---|---|---|---|
| 5/19/2023 14:56 | 176.93 | 220.48 | 287.2 | 241.32 | 250.48 | 250.76 |
| 5/19/2023 14:57 | 178.29 | 223.69 | 288.33 | 244.84 | 254.33 | 254.03 |
| 5/19/2023 14:58 | 179.16 | 222.67 | 289.72 | 246.87 | 256.79 | 256.77 |
| 5/19/2023 14:59 | 181.17 | 226.64 | 293.48 | 250.46 | 261.49 | 259.81 |
| 5/19/2023 15:00 | 182.47 | 227.48 | 293.19 | 251.68 | 262.87 | 262.6 |
| 5/19/2023 15:01 | 184.44 | 230.4 | 294.21 | 252.92 | 264.62 | 265.37 |
| 5/19/2023 15:02 | 184.81 | 231.5 | 294.27 | 252.49 | 268.55 | 266.92 |
| 5/19/2023 15:03 | 186.09 | 234.2 | 295.07 | 253.96 | 268.86 | 269.25 |
| 5/19/2023 15:04 | 185.92 | 237.18 | 398.86 | 256.51 | 273.05 | 271.66 |
| 5/19/2023 15:05 | 187.59 | 240.7 | 301.43 | 259.21 | 274.07 | 272.51 |
| 5/19/2023 15:06 | 187.46 | 236.85 | 300.39 | 259.83 | 274.03 | 274.96 |
| 5/19/2023 15:07 | 189.76 | 243.52 | 302.27 | 259.27 | 275.4 | 275.27 |
| 5/19/2023 15:08 | 188.76 | 241.65 | 303.3 | 262.4 | 278.52 | 277.06 |
| 5/19/2023 15:09 | 190.31 | 246.22 | 304.75 | 262.2 | 277.44 | 276.84 |
| 5/19/2023 15:10 | 189.75 | 241.69 | 305.32 | 264.92 | 279.75 | 279.6 |
| 5/19/2023 15:11 | 190.03 | 243.28 | 306.67 | 264.31 | 281.4 | 279.99 |
| 5/19/2023 15:12 | 189.78 | 244.78 | 306.51 | 265.27 | 281.53 | 281.39 |
| 5/19/2023 15:13 | 189.72 | 245.48 | 307.97 | 265.61 | 283.48 | 281.96 |
| 5/19/2023 15:14 | 190.81 | 245.03 | 307.11 | 264.35 | 284.56 | 284.15 |
| 5/19/2023 15:15 | 190.98 | 246.93 | 309.22 | 266.07 | 285.1 | 283.79 |
| 5/19/2023 15:16 | 191.26 | 246.44 | 309.42 | 265.15 | 282.55 | 284.39 |
| 5/19/2023 15:17 | 192.49 | 247.5 | 308.8 | 267.64 | 285.92 | 285.37 |
| 5/19/2023 15:18 | 193.3 | 249.21 | 307.4 | 267.52 | 284.05 | 285.54 |
| 5/19/2023 15:19 | 193.59 | 244.61 | 306.55 | 265.15 | 284.93 | 285.38 |
| 5/19/2023 15:20 | 195.17 | 245.9 | 306.69 | 267.41 | 283.35 | 285.44 |
| 5/19/2023 15:21 | 195.41 | 245.47 | 305.31 | 265.02 | 283.44 | 285.55 |
| 5/19/2023 15:22 | 193.95 | 245.15 | 308.36 | 267.73 | 285.35 | 286.4 |
| 5/19/2023 15:23 | 193.99 | 246.04 | 309.7 | 268.56 | 285.77 | 285.2 |
| 5/19/2023 15:24 | 194.26 | 247.77 | 309.15 | 269.45 | 285.45 | 284.77 |
| 5/19/2023 15:25 | 194.59 | 246.9 | 310 | 269.34 | 283.47 | 286.35 |
| 5/19/2023 15:26 | 194.03 | 245.96 | 308.49 | 267.97 | 282.74 | 285.27 |
| 5/19/2023 15:27 | 193.15 | 248.6 | 309.93 | 268.96 | 283.89 | 286.64 |
| 5/19/2023 15:28 | 194.65 | 245.95 | 309.78 | 267.86 | 283.89 | 286.18 |
| 5/19/2023 15:29 | 194.01 | 248.53 | 310.35 | 269.51 | 287.21 | 286.8 |
| 5/19/2023 15:30 | 194.93 | 250.89 | 311.42 | 269.21 | 287.05 | 287.15 |
| 5/19/2023 15:31 | 195.51 | 249.7 | 311.99 | 269.19 | 287.81 | 288.24 |
| 5/19/2023 15:32 | 195.59 | 253.14 | 312.59 | 269.3 | 288.06 | 288.02 |
| 5/19/2023 15:33 | 196.34 | 251.59 | 312.09 | 270.13 | 288.01 | 287.78 |
| 5/19/2023 15:34 | 195.75 | 249.53 | 312.01 | 270.02 | 288.02 | 289.1 |

FIG. 18

BURNING ON MAXIMUM FLAME;
TEST TIME: 1 HOUR.
ENVIRONMENT TEMPERATURE IS 29.6C

| SINGLE PANE DOMAIN | | | |
|---|---|---|---|
| GLASS PANE# | LOCATION | ACTUAL(F) | TEMP. SPEC |
| 1 (OUTER) | TOP | 196.27 | GLASS SURFACE TEMPERATURE DOES NOT EXCEED 82.6C (180.68F) |
| 1 (OUTER) | BOTTOM | 249.95 | |
| 2 (OUTER) | TOP | 312.24 | |
| 2 (OUTER) | BOTTOM | 270.52 | |
| 3 (OUTER) | TOP | 284.59 | |
| 3 (OUTER) | BOTTOM | 289.45 | |

FIG. 19

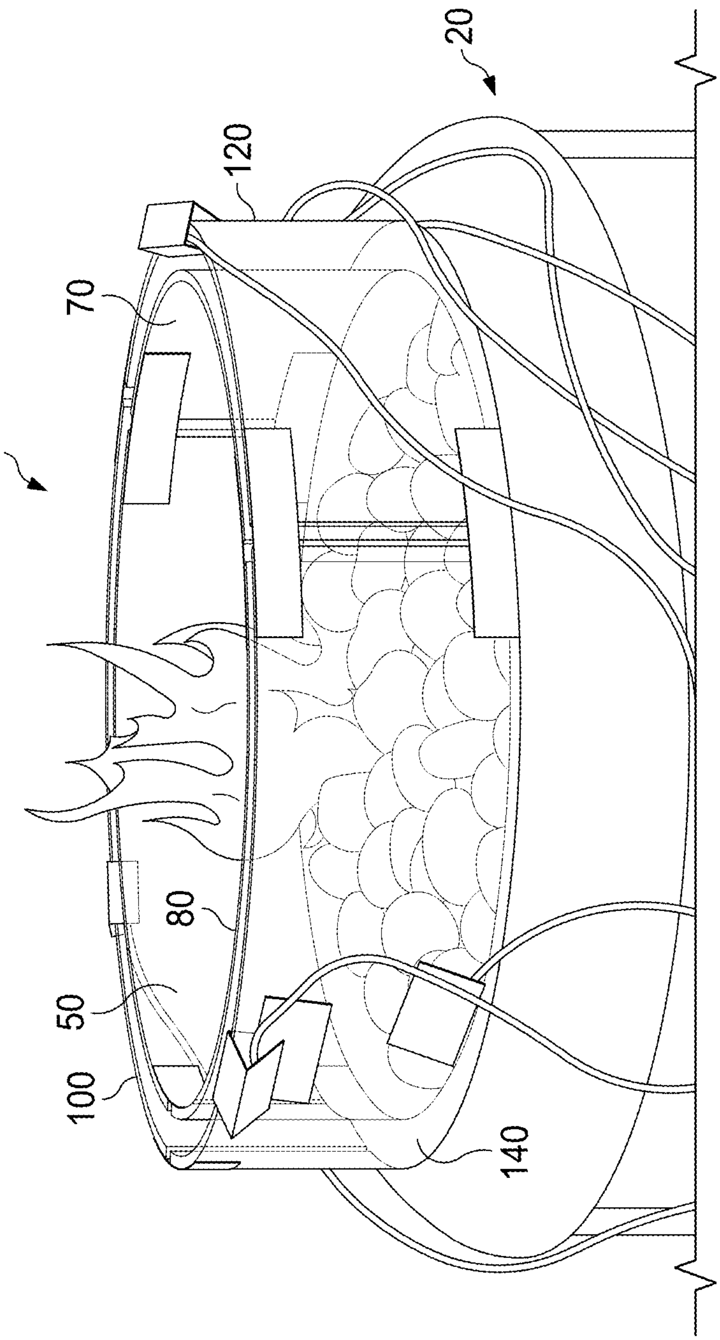


FIG. 20

| TIME | TOP 1 | BOTTOM 1 | TOP 2 | BOTTOM 2 | TOP 3 | BOTTOM 3 |
|---|---|---|---|---|---|---|
| 2023/5/30 14:15 | 113.43 | 118.41 | 141.64 | 123.13 | 132.45 | 122.86 |
| 2023/5/30 14:16 | 113.83 | 118.6 | 143.46 | 124.32 | 133.5 | 124.31 |
| 2023/5/30 14:17 | 114.29 | 119.34 | 144.13 | 125.27 | 134.04 | 124.62 |
| 2023/5/30 14:18 | 114.39 | 120.63 | 145.36 | 125.9 | 135.42 | 126 |
| 2023/5/30 14:19 | 115.1 | 120.73 | 145.97 | 126.71 | 136.17 | 126.53 |
| 2023/5/30 14:20 | 116.47 | 121.65 | 146.07 | 127.18 | 137.09 | 126.89 |
| 2023/5/30 14:21 | 117.14 | 122.37 | 147 | 128.23 | 137.74 | 127.13 |
| 2023/5/30 14:22 | 117.12 | 122.85 | 148.77 | 128.39 | 138.38 | 128.1 |
| 2023/5/30 14:23 | 117.5 | 123.24 | 151.39 | 129.97 | 138.68 | 128.66 |
| 2023/5/30 14:24 | 116.85 | 123.78 | 150.49 | 130.24 | 139.48 | 129.52 |
| 2023/5/30 14:25 | 117.33 | 124.33 | 151 | 131.23 | 140.37 | 130.34 |
| 2023/5/30 14:26 | 117.62 | 125.68 | 150.71 | 131.09 | 140.53 | 130.59 |
| 2023/5/30 14:27 | 117.62 | 125.39 | 151.13 | 132.17 | 140.6 | 131.18 |
| 2023/5/30 14:28 | 118.37 | 125.94 | 151.58 | 132.3 | 141.6 | 131.28 |
| 2023/5/30 14:29 | 118.43 | 126.44 | 152.54 | 133.18 | 142.2 | 132.16 |
| 2023/5/30 14:30 | 118.8 | 126.09 | 156.17 | 134.07 | 141.08 | 131.87 |
| 2023/5/30 14:31 | 118.7 | 126.27 | 154.75 | 133.67 | 141.36 | 132.45 |
| 2023/5/30 14:32 | 118.51 | 126.92 | 154.94 | 134.19 | 141.92 | 132.39 |
| 2023/5/30 14:33 | 118.64 | 127.09 | 154.86 | 134.42 | 142.14 | 132.94 |
| 2023/5/30 14:34 | 118.94 | 126.32 | 155.33 | 134.11 | 142.33 | 132.91 |
| 2023/5/30 14:35 | 118.73 | 127.33 | 155.09 | 134.59 | 142.51 | 132.74 |
| 2023/5/30 14:36 | 119.85 | 126.83 | 155.23 | 134.92 | 142.38 | 132.9 |
| 2023/5/30 14:37 | 119.62 | 128.27 | 155.12 | 135.48 | 143.4 | 133.75 |
| 2023/5/30 14:38 | 119.65 | 127.5 | 156.08 | 135.11 | 143.82 | 133.58 |
| 2023/5/30 14:39 | 119.19 | 127.77 | 157.05 | 135.97 | 144.08 | 134.41 |
| 2023/5/30 14:40 | 119.89 | 128.5 | 157.76 | 135.94 | 142.83 | 133.64 |
| 2023/5/30 14:41 | 119.84 | 127.45 | 157.68 | 135.7 | 142.83 | 133.93 |
| 2023/5/30 14:42 | 119.74 | 128.37 | 155.81 | 134.98 | 143.44 | 134.09 |
| 2023/5/30 14:43 | 120.07 | 127.69 | 156.44 | 135.71 | 143.18 | 133.93 |
| 2023/5/30 14:44 | 119.67 | 128.58 | 156.45 | 136.25 | 143.48 | 134.29 |
| 2023/5/30 14:45 | 120.29 | 127.91 | 155.94 | 135.54 | 143.87 | 133.97 |
| 2023/5/30 14:46 | 119.86 | 128.48 | 155.57 | 135.39 | 143.83 | 134.23 |
| 2023/5/30 14:47 | 120.4 | 128.76 | 156.82 | 136.12 | 143.66 | 134.26 |
| 2023/5/30 14:48 | 120.21 | 128.74 | 157.65 | 136.82 | 143.59 | 134.5 |
| 2023/5/30 14:49 | 120.48 | 128.98 | 156.19 | 136.45 | 143.99 | 134.87 |

FIG. 21

BURNING ON MAXIMUM FLAME;
TEST TIME: 1 HOUR.
ENVIRONMENT TEMPERATURE IS 30.9C

| DOUBLE PANE DOMAIN | | | |
|---|---|---|---|
| GLASS PANE# | LOCATION | ACTUAL(F) | TEMP. SPEC |
| 1 (OUTER) | TOP | 120.48 | GLASS SURFACE TEMPERATURE DOES NOT EXCEED 83.9C (183.02F) |
| 1 (OUTER) | BOTTOM | 128.98 | |
| 2 (OUTER) | TOP | 156.19 | |
| 2 (OUTER) | BOTTOM | 136.45 | |
| 3 (OUTER) | TOP | 143.99 | |
| 3 (OUTER) | BOTTOM | 134.87 | |

FIG. 22

# MULTI-PANE PORTABLE GLASS SURROUND FOR PROPANE FIRE PIT

## CROSS-REFERENCE TO RELATED CASES

**[0001]** This application claims the benefit of U.S. provisional patent application Ser. No. 63/563,046, filed on Mar. 8, 2024, and incorporates such provisional application by reference into this disclosure as if fully set out at this point.

## FIELD OF THE INVENTION

**[0002]** The invention relates to heat shielding in general and, more particularly, to a multi-pane glass surround for a fire pit.

## BACKGROUND OF THE INVENTION

**[0003]** For performance, appearance, and other reasons, open flames from fire pits may be surrounded or shielded when the fire pit is in operation. When an opaque shield is utilized, the desired heat effects may be achieved, but at the expense of loss of desired appearance and lighting. Prior transparent or translucent shielding suffers from poor performance with respect to heat control.

**[0004]** What is needed is a system and method for addressing the above and related issues.

## SUMMARY OF THE INVENTION

**[0005]** The invention of the present disclosure, in one aspect thereof, comprises a fire pit surround having an inner wall and an outer wall surrounding the inner wall. The inner wall and outer wall are spaced apart from one another and define a central opening within the inner wall for flame from a fire pit to pass into.

**[0006]** The inner and outer wall may be spaced apart to allow airflow therebetween. The inner and outer walls may be cylindrical and may be coaxial with respect to one another.

**[0007]** The inner wall and outer wall may comprise a plurality of inner and outer panes, respectively. In some cases, the plurality of inner panes are spaced apart from one another to define a plurality of inner gaps therebetween, and the outer panes are spaced apart from one another to define a plurality of outer gaps therebetween.

**[0008]** The plurality of inner panes and the plurality of outer panes may have a cross section defining a circle arc. At least one of the plurality of inner gaps can be circumferential aligned with at least one of the plurality of outer gaps. In some cases, each of the plurality of inner gaps is circumferentially aligned with one of the plurality of outer gaps.

**[0009]** The fire pit may include at least one bracket joining at least first and second inner panes of the plurality of inner panes and at least first and second outer panes of the plurality of outer panes. The bracket may define an inside vertical member having a first width separating the first and second inner pane, and an outside vertical member having a second width separating the first and second outer pane. The bracket may provide inner surfaces adjacent the inside vertical member for receiving the first and second inner pane and provides outer surfaces adjacent the outside vertical member for receiving the first and second outer panes.

**[0010]** The bracket may join the first inner pane and the second inner pane such that the first inner pane has a bottom edge that is below a bottom edge of the first outer pane. The bracket may provide a pair of support surfaces adjacent the outside vertical member that supports respective the first and second outer panes.

**[0011]** The invention of the present disclosure, in another aspect thereof, comprises a fire pit surround having a first upright inner cylindrical wall sized to circumscribe a burner of a fire pit, and a second upright outer cylindrical wall spaced apart from and surrounding the first upright inner wall. The first upright inner wall and the second upright inner wall comprise a plurality of inner and outer panes, respectively, the inner and outer panes being spaced apart to define inner and outer gaps, respectively, the inner and outer gaps being aligned circumferentially.

**[0012]** The fire pit may have at least one bracket joining to at least first and second inner panes of the plurality of inner panes and at least first and second outer panes of the plurality of outer panes, the bracket maintaining the first and second inner panes and the first and second outer panes in a spaced apart relationship. In some cases, the bracket defines inner and outer vertical members between the first and second inner panes and first and second outer panes, respectively. The bracket may join the first and second inner panes such that respective first and second bottom edges of the first and second inner panes have a different elevation from respective first and second bottom edges of the first and second outer panes.

**[0013]** In some cases, the bracket provides first and second support surfaces adjacent the outer vertical member for receiving the first and second bottom edges of the first and second outer panes, respectively. The bracket may define a first oval orifice receiving a first fastener passing through the first inner pane and the first outer pane, and may define a second oval orifice receiving second fastener passing through the second inner pane and the second outer pane.

## BRIEF DESCRIPTION OF THE DRAWINGS

**[0014]** FIG. **1** is a perspective view of a fire pit surround according to aspects of the present disclosure.

**[0015]** FIG. **2** is a perspective view of the fire pit surround of FIG. **1** mounted on a propane burning fire pit.

**[0016]** FIG. **3** is a perspective view of a first inner pane of the fire pit surround of FIG. **1**.

**[0017]** FIG. **4** is a perspective view of a first outer of the fire pit surround of FIG. **1**.

**[0018]** FIG. **5** is a perspective view of a second inner pane of the fire pit surround of FIG. **1**.

**[0019]** FIG. **6** is a perspective view of a second outer pane of the fire pit surround of FIG. **1**.

**[0020]** FIG. **7**A is a perspective view of a bracket engaging panes of the fire pit surround of FIG. **1**.

**[0021]** FIG. **7**B is a plan view of a bracket of the fire pit surround of FIG. **1**.

**[0022]** FIG. **8** is an elevation view of a bracket of the fire pit surround of FIG. **1**.

**[0023]** FIG. **9** is another perspective view of the fire pit surround of FIG. **1**.

**[0024]** FIG. **10** is a perspective view of the fire pit surround of FIG. **1** partially assembled.

**[0025]** FIG. **11** is a partially exploded perspective view of the fire pit surround of FIG. **1** showing fastener locations.

**[0026]** FIG. **12** is another perspective view of the fire pit surround of FIG. **1**.

**[0027]** FIG. **13** is a perspective view of a fire pit surround according to the present disclosure mounted on an operational gas fire pit.

**[0028]** FIG. **14** is a perspective view of a fire pit surround according to the present disclosure mounted on an operational table top fire pit.

**[0029]** FIG. **15** is a perspective view of a fire pit surround of the present disclosure mounted on an outdoor heater.

**[0030]** FIG. **16** is a perspective view of a fire pit surround of the present disclosure mounted on an outdoor table top heater.

**[0031]** FIG. **17** is a perspective view of a single pane fire pit surround mounted on a table top fire pit for comparison testing.

**[0032]** FIG. **18** is a table of experimental data from the testing of FIG. **17**.

**[0033]** FIG. **19** is a summary table of experimental data from the testing of FIG. **17**.

**[0034]** FIG. **20** is a perspective view of a double pane fire pit surround according to the present disclosure mounted on a table top fire pit for comparison testing.

**[0035]** FIG. **21** is a table of experimental data from the testing of FIG. **20**.

**[0036]** FIG. **22** is a summary table of experimental data from the testing of FIG. **20**.

## DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

**[0037]** Referring now to FIG. **1**, a perspective view of a fire pit surround **10** according to aspects of the present disclosure is shown. Referring also to FIG. **2**, a perspective view of the fire pit surround **10** of FIG. **1** mounted on a propane burning fire pit **20** is shown. Various embodiments of the present disclosure, such as fire pit surround **10**, provide a double walled barrier **12** around a heat source, such as a flame of a fire pit. The double walled barrier **12** may define an open top and bottom allowing the surround **10** to fit onto or sit on a portion of a fire pit or other flame producing device. The burner (or burners) or flames produced by such device will thus be surrounded and shielded by the surround **10** in general and the barrier **12** in particular. While embodiments of the present disclosure may be particularly suited to use with gas or propane burning devices, they may also be adapted for use with devices burning solid fuels.

**[0038]** The double walled barrier **12** may comprise an inner wall **40** surrounded and spaced apart from an outer wall **90**. The inner wall **40** and the outer wall **90** may be coaxially spaced apart around a common center such that a distance between the inner wall **40** and outer wall **90** remains constant or substantially so around the circumference. The barrier **12** may be generally cylindrical, or circular in horizontal cross section. In other embodiments, the barrier **12** may ascend at an angle. In some embodiments, the inner wall **40** may have a different upright angle than the outer wall **90**.

**[0039]** The inner wall **40** may comprise a plurality of inner panes **50**, **70**, **80**. As shown the inner wall **40** comprises three inner panes **50**, **70**, **80**, but more or fewer could be utilized. Each pane **50**, **70**, **80** may be curved and may form an arc of a circle (in horizontal cross section). The panes **50**, **70**, **80** may be joined together by brackets **150**, discussed further below.

**[0040]** The outer wall **90** may comprise a plurality of outer panels **100**, **120**, **140**. As shown, the outer wall **90** comprises three outer panes **100**, **120**, **140** but more or fewer could be utilized. Each pane **50**, **70**, **80** may be curved, and may form an arc of a circle (in horizontal cross section). The panes **50**, **70**, **80** may be joined together by brackets **150**, discussed further below.

**[0041]** While the inner wall **40** is spaced apart from the outer wall **90**, the inner panes **50**, **70**, **80** of the inner wall **40** may have spaces or gaps **14** therebetween. The outer panes **100**, **120**, **140** of the outer wall **90** may have spaces or gaps **16** therebetween as well. Such spaces or gaps **14**, **16** may extend the height of the respective wall **40**, **90**. The spaces or gaps **14**, **16** between the inner panes **50**, **70**, **80** and outer panels **100**, **120**, **140**, respectively, may be aligned. In other embodiments such spaces or gaps **14**, **16** are offset circumferentially.

**[0042]** In some embodiments, the inner wall **40** may have a height that differs from a height of the outer wall **90**. In some cases, the inner wall **40** extends below the outer wall **90**. This may enable the surround **10** to fit into a wall, lip, or receptacle **18** of fire pit **20** (or other flame or heat producing device). This may serve to properly locate or secure the surround **10** into a useful operational position. In some embodiments, the brackets **150** may be at the top of the respective walls **40**, **90** and the inner wall **40** and outer wall **90** may be fitted into a stationary receiving bracket **22** on the fire pit **20** (or other flame or heat producing device).

**[0043]** The walls **40**, **90** and the respective panels **50**, **70**, **80** and **100**, **120**, **140** may comprise glass. Tempered glass or heat treated glass may be used. The glass may be made heat resistant by means known to the art.

**[0044]** The walls **40**, **90** and associated components are illustrated as being configured in a circular or cylindrical arrangement. Such arrangement may be convenient for circular fire pits, heaters, and other devices. However, it will be appreciated that by modification of the brackets **150** and the shape of the walls **40**, **90** generally, and the panes **50**, **70**, **80**, **100**, **120**, **140** specifically, the surround **10** can be configured in a square shape, a rectangular shape, an oval shape, or whatever geometric configuration most appropriately fits the flame or heat producing device with which the surround is utilized.

**[0045]** Referring now to FIG. **3**, a perspective view of a first inner pane **50** of the fire pit surround **10** of FIG. **1** is shown. Inner pane **50** may have inner surface **52**, outer surface **54**, right edge **56**, top edge **58**, and bottom edge **60**. First inner pane **50** may define a first inner attachment orifice **62**. First inner attachment orifice **62** may be proximate to right edge **56** of first inner pane **50** and may be located at a desired distance from top edge **58** or bottom edge **60** of first inner pane **50**. A second inner attachment orifice **63** may be proximate a left edge **57** of first inner pane **50** and may be located at a desired distance from top edge **58** or bottom edge **60** of inner pane **50**.

**[0046]** Referring now to FIG. **5**, a perspective view of a inner pane **70** of the fire pit surround **10** of FIG. **1** is shown. Inner pane **70** may have inner surface **72**, outer surface **74**, left edge **76**, top edge **78**, bottom edge **80**. The inner pane **70** may define a first inner attachment orifice **82** proximate to left edge **76** of second inner pane **70**. First inner attachment orifice **82** may additionally be located a desired distance from top edge **78** or bottom edge **80** of inner pane **70**. The inner pane **70** may define a second inner attachment orifice

**83** proximate a right edge **77** of second inner pane **70**. Second inner attachment orifice **83** may additionally be located a desired distance from top edge **78** or bottom edge **80** of inner pane **70**.

**[0047]** Referring now to FIG. **4**, a perspective view of a outer pane **100** of the fire pit surround **10** of FIG. **1** is shown. Outer pane **100** may have an inner surface **102**, outer surface **104**, right edge **106**, top edge **108**, and bottom edge **110**. Outer pane **100** may define a first outer attachment orifice **112**. First curved attachment orifice **112** may be proximate to a corner defined by right edge **106** and either top edge **108** or bottom edge **110**. Outer pane **100** may define a second outer attachment orifice **113**. Second outer attachment orifice **113** may be proximate to a corner defined by left edge **107** and either top edge **108** or bottom edge **110**.

**[0048]** Referring now to FIG. **6**, a perspective view of a outer pane **120** of the fire pit surround **10** of FIG. **1** is shown. Outer pane **120** may have inner surface **122**, outer surface **124**, left edge **126**, top edge **128**, and bottom edge **130**. Outer pane **120** may define first outer attachment orifice **132**. First outer attachment orifice **132** may be proximate to a corner defined by left edge **126** and either top edge **128** or bottom edge **130**. The outer pane **120** may define second outer attachment orifice **133**. Second outer attachment orifice **133** may be proximate to a corner defined by right edge **127** and either top edge **128** or bottom edge **130**.

**[0049]** It should be appreciated that inner pane **80** may have an identical or substantially similar construction as inner panes **50**, **70** described above. It should also be appreciated that outer pane **140** may have an identical or substantially similar construction as outer panes **100**, **120** described above.

**[0050]** Referring now to FIG. **7**A a perspective view of a bracket **150** engaging panes **50**, **70**, **120** of the fire pit surround **10** of FIG. **1** is shown. In an embodiment, one of brackets **150** is affixed to first inner pane **50**, second inner pane **70**, first outer pane **100**, and second outer pane **120** (for clarity, pane **100** not shown in FIG. **7**A). Referring also to FIG. **7**B, a plan view of a bracket **150** of the fire pit surround **10** of FIG. **1** is shown. Referring also to FIG. **8**, an elevation view of a bracket **150** of the fire pit surround **10** of FIG. **1** is shown.

**[0051]** Brackets **150** may define right inner groove **160**, left inner groove **162**, right outer groove **164**, and left outer groove **166**. Second, third, and/or subsequent ones of brackets **150** may be constructed identically or substantially similarly to the described bracket **150**.

**[0052]** As shown in FIGS. **1**, **2**, **9**, and **12**, for example, in embodiments of glass surround **10** having inner wall **40** comprised of first inner pane **50**, second inner pane **70**, and third inner pane **80** and having an outer wall **90** comprised of first outer pane **100**, second outer pane **120**, and third outer pane **140**, one of brackets **150** is affixed to second inner pane **70**, third inner pane **80**, second outer pane **120**, and third outer pane **140**. A second one of brackets **150** is affixed to third inner pane **80**, first inner pane **50**, third outer panel **140**, and first outer pane **100**. It should be understood that, depending on the number of panes that comprise inner wall **40** and outer wall **90**, a number of brackets **150** will be provided to interconnect adjacent panes.

**[0053]** Still referring to FIGS. **7**A, **7**B, and **8**, bracket **150** may have outside surface **168**, inside surface **170**, and body portion **172**. Outside surface **168** may define horizontal member **174**, which may define support surface **176** for supporting first outer pane **100** and second outer pane **120**. Horizontal member **174** may have a left portion and right portion that extend from either side of outside vertical member **182**, which may have left outside surface and right outside surface. Outside vertical member **182** may define a width **183**.

**[0054]** Inside surface **170** of first bracket **150** may define inside vertical member **188** having first inside surface **190** and a second inside surface **192**. Inside vertical member **188** may define a width.

**[0055]** Body portion **172** of bracket **150** may have an outside surface **196** and inside surface **198**. Outside surface **196** of body portion **172** may have first outer surface **200** located on a first side of outside vertical member **182** and above a left portion of horizontal member **174**. Outside surface **196** may have a second outer surface **202** located to a second side of outside vertical member **182** and above a right portion of horizontal member **174**.

**[0056]** Inside surface **198** of middle portion **172** may have a first inside surface **204** located adjacent a first side of inside vertical member **188** and may have a second inside surface **206** located adjacent a second side of inside vertical member **188**. Body portion **172** may define first bracket orifice **208** that passes from first outer surface **200** to first inside surface **204**. Body portion **172** may define second bracket orifice **210** that passes from second outer surface **202** to second inside surface **206**.

**[0057]** Bottom edge **110** of first outer pane **100** may be received adjacent to support surface **176** of the left portion of horizontal member **174**. Right edge **106** of first outer pane **100** may be adjacent to first left outer groove **166** of bracket **150**. Bottom edge **130** of second outer pane **120** may be received adjacent to support surface **176** of a right portion of horizontal member **174**. Left edge **126** of second outer pane **120** may be adjacent to right outer groove **164** of bracket **150**.

**[0058]** As best seen in FIG. **11**, a first fastener **220** may be received in first outer attachment orifice **112** of outer pane **100** and pass through first bracket orifice **208** for securing pane **100** to first outer surface **200** and of first bracket **150**. First fastener **220** may pass through first inner attachment orifice **62** of inner pane **50** for securing first inner pane **50** to first inside section **190** of bracket **150**.

**[0059]** As best seen in FIG. **10**, a second fastener **230** may be received in second outer attachment orifice **132** of outer pane **120** and pass through bracket orifice **210** for securing outer pane **120** to second outer surface **202** of first bracket **150**. Second fastener **230** may pass through second attachment orifice **82** of inner pane **70** for securing second inner pane **70** to second inner surface **192** of first bracket **150**.

**[0060]** In some embodiments, some or all of the various orifices defined in the brackets **150**, the inner panes **50**, **70**, **80**, and the outer panes **100**, **120**, **140** may not be perfectly circular. Some or all of the orifices may define an oval shape for ease of assembly.

**[0061]** FIG. **12** provides a completed view of the surround **10** in an inverted configuration following assembly (as in FIGS. **10-11**) but prior to installation on a fire pit or other device.

**[0062]** As best seen in FIG. **7**A, right edge **56** of inner pane **50** and left edge **76** of inner pane **70** may define inside gap **240** therebetween. Inside edge gap **240** may have a width that is approximately the same dimension as width **194** of inside vertical member **188**. As best seen in FIG. **9**, right edge **106** of outer pane **100** and left edge **126** of outer pane **120** may define outside edge gap **250** therebetween. Outside edge gap **250** may have a width **252** that is approximately the same dimension as width **183** of outside vertical member **182**.

**[0063]** Outer surface **54** of inner pane **50** and inner surface **102** of outer pane **100** may define an air gap **260** therebetween. Air gap **260** may have has a width **262** that is approximately the same dimension as the distance between first inside surface **190** and first outer surface **200** of first bracket **150**. Outer surface **74** of inner pane **70** and inner surface **122** of outer pane **120** may also define air gap **260** therebetween.

**[0064]** The bottom edge **60** of inner pane **50** and bottom edge **80** of inner pane **70** may extend below bottom edge **110** of first outer pane **100** and bottom edge **130** of outer pane **120** since bottom edge **60** of first inner pane **50** and bottom edge **80** of second inner pane **70** may not engage support surfaces and are therefore able to be positioned a desired distance above or below bottom edges **110**, **130** of outer panes **100**, **120**.

**[0065]** Referring now to FIG. **13**, a perspective view of a fire pit surround **10** according to the present disclosure mounted on an operational gas fire pit **20** is shown. Referring to FIG. **14**, a perspective view of a fire pit surround **10** according to the present disclosure mounted on an operational table top fire pit **22**. FIG. **15** provides a perspective view of a fire pit surround **10** of the present disclosure mounted on an outdoor heater **23**. FIG. **16** is a perspective view of a fire pit surround **10** of the present disclosure mounted on an outdoor table top heater **24**.

**[0066]** Referring now to FIG. **17**, a perspective view of a single pane fire pit surround **1700** mounted on a table top fire pit **22** for comparison testing. FIG. **18** is a table of experimental data from the testing of FIG. **17**. FIG. **18** shows temperature probe measurement at the top and bottom of the surround **1700** as the fire pit **22** was operational. FIG. **19** is a summary table of experimental data from the testing of FIG. **17**. Here it can be seen that over a one hour test, the temperature at every measured location exceeded a specification temperature of 180.68° F.

**[0067]** FIG. **20** is a perspective view of a double pane fire pit surround **10** according to the present disclosure mounted on a table top fire pit **20** for comparison testing. FIG. **21** is a table of experimental data from the testing of FIG. **20**. FIG. **22** is a summary table of experimental data from the testing of FIG. **20**. Here it can be seen that over a one hour test, the temperature at every measured location did not exceed 156.19° F. in the worst case, which was below the 183° F. (83.9° C.) specification temperature. Thus the surround **10** of the present disclosure performs markedly better than a single pane system.

**[0068]** It is to be understood that the terms "including", "comprising", "consisting" and grammatical variants thereof do not preclude the addition of one or more components, features, steps, or integers or groups thereof and that the terms are to be construed as specifying components, features, steps or integers.

**[0069]** If the specification or claims refer to "an additional" element, that does not preclude there being more than one of the additional element.

**[0070]** It is to be understood that where the claims or specification refer to "a" or "an" element, such reference is not be construed that there is only one of that element.

**[0071]** It is to be understood that where the specification states that a component, feature, structure, or characteristic "may", "might", "can" or "could" be included, that particular component, feature, structure, or characteristic is not required to be included.

**[0072]** Where applicable, although state diagrams, flow diagrams or both may be used to describe embodiments, the invention is not limited to those diagrams or to the corresponding descriptions. For example, flow need not move through each illustrated box or state, or in exactly the same order as illustrated and described.

**[0073]** Methods of the present invention may be implemented by performing or completing manually, automatically, or a combination thereof, selected steps or tasks.

**[0074]** The term "method" may refer to manners, means, techniques and procedures for accomplishing a given task including, but not limited to, those manners, means, techniques and procedures either known to, or readily developed from known manners, means, techniques and procedures by practitioners of the art to which the invention belongs.

**[0075]** The term "at least" followed by a number is used herein to denote the start of a range beginning with that number (which may be a range having an upper limit or no upper limit, depending on the variable being defined). For example, "at least 1" means 1 or more than 1. The term "at most" followed by a number is used herein to denote the end of a range ending with that number (which may be a range having 1 or 0 as its lower limit, or a range having no lower limit, depending upon the variable being defined). For example, "at most 4" means 4 or less than 4, and "at most 40%" means 40% or less than 40%.

**[0076]** When, in this document, a range is given as "(a first number) to (a second number)" or "(a first number)-(a second number)", this means a range whose lower limit is the first number and whose upper limit is the second number. For example, 25 to 100 should be interpreted to mean a range whose lower limit is 25 and whose upper limit is 100. Additionally, it should be noted that where a range is given, every possible subrange or interval within that range is also specifically intended unless the context indicates to the contrary. For example, if the specification indicates a range of 25 to 100 such range is also intended to include subranges such as 26-100, 27-100, etc., 25-99, 25-98, etc., as well as any other possible combination of lower and upper values within the stated range, e.g., 33-47, 60-97, 41-45, 28-96, etc. Note that integer range values have been used in this paragraph for purposes of illustration only and decimal and fractional values (e.g., 46.7-91.3) should also be understood to be intended as possible subrange endpoints unless specifically excluded.

**[0077]** It should be noted that where reference is made herein to a method comprising two or more defined steps, the defined steps can be carried out in any order or simultaneously (except where context excludes that possibility), and the method can also include one or more other steps which are carried out before any of the defined steps, between two of the defined steps, or after all of the defined steps (except where context excludes that possibility).

**[0078]** Further, it should be noted that terms of approximation (e.g., "about", "substantially", "approximately", etc.) are to be interpreted according to their ordinary and customary meanings as used in the associated art unless indicated otherwise herein. Absent a specific definition within this disclosure, and absent ordinary and customary usage in the associated art, such terms should be interpreted to be plus or minus 10% of the base value.

**[0079]** Thus, the present invention is well adapted to carry out the objects and attain the ends and advantages mentioned above as well as those inherent therein. While the inventive device has been described and illustrated herein by reference to certain preferred embodiments in relation to the drawings attached thereto, various changes and further modifications, apart from those shown or suggested herein, may be made therein by those of ordinary skill in the art, without departing from the spirit of the inventive concept the scope of which is to be determined by the following claims.

What is claimed is:

**1**. A fire pit surround comprising:

an inner wall;

an outer wall surrounding the inner wall;

wherein the inner wall and outer wall are spaced apart from one another and define a central opening within the inner wall for flame from a fire pit to pass into.

**2**. The fire pit surround of claim **1**, wherein the inner and outer wall are spaced apart to allow airflow therebetween.

**3**. The fire pit surround of claim **2**, wherein the inner and outer walls are cylindrical.

**4**. The fire pit surround of claim **3**, wherein the inner and outer walls are coaxial with respect to one another.

**5**. The fire pit surround of claim **1**, wherein the inner wall and outer wall comprise a plurality of inner and outer panes, respectively.

**6**. The fire pit surround of claim **1**:

wherein the plurality of inner panes are spaced apart from one another to define a plurality of inner gaps therebetween; and

wherein the outer panes are spaced apart from one another to define a plurality of outer gaps therebetween.

**7**. The fire pit surround of claim **6**, wherein the plurality of inner panes and the plurality of outer panes have a cross section defining a circle arc.

**8**. The fire pit surround of claim **7**, wherein at least one of the plurality of inner gaps is circumferential aligned with at least one of the plurality of outer gaps.

**9**. The fire pit surround of claim **8**, wherein each of the plurality of inner gaps is circumferentially aligned with one of the plurality of outer gaps.

**10**. The fire pit surround of claim **6**, further comprising at least one bracket joining at least first and second inner panes of the plurality of inner panes and at least first and second outer panes of the plurality of outer panes.

**11**. The fire pit surround of claim **10**, wherein the at least one bracket defines:

an inside vertical member having a first width separating the first and second inner pane; and

an outside vertical member having a second width separating the first and second outer pane.

**12**. The fire pit surround of claim **11**, wherein the at least one bracket provides inner surfaces adjacent the inside vertical member for receiving the first and second inner pane and provides outer surfaces adjacent the outside vertical member for receiving the first and second outer panes.

**13**. The fire pit surround of claim **12**, wherein the at least one bracket joins the first inner pane and the second inner pane such that the first inner pane has a bottom edge that is below a bottom edge of the first outer pane.

**14**. The fire pit surround of claim **13**, wherein the at least one bracket provides a pair of support surfaces adjacent the outside vertical member that supports respective the first and second outer panes.

**15**. A fire pit surround comprising:

a first upright inner cylindrical wall sized to circumscribe a burner of a fire pit; and

a second upright outer cylindrical wall spaced apart from and surrounding the first upright inner wall;

wherein the first upright inner wall and the second upright inner wall comprise a plurality of inner and outer panes, respectively, the inner and outer panes being spaced apart to define inner and outer gaps, respectively, the inner and outer gaps being aligned circumferentially.

**16**. The fire pit surround of claim **15**, further comprising at least one bracket joining to at least first and second inner panes of the plurality of inner panes and at least first and second outer panes of the plurality of outer panes, the bracket maintaining the first and second inner panes and the first and second outer panes in a spaced apart relationship.

**17**. The fire pit surround of claim **16**, wherein the at least one bracket defines inner and outer vertical members between the first and second inner panes and first and second outer panes, respectively.

**18**. The fire pit surround of claim **17**, wherein the at least one bracket joins the first and second inner panes such that respective first and second bottom edges of the first and second inner panes have a different elevation from respective first and second bottom edges of the first and second outer panes.

**19**. The fire pit surround of claim **18**, wherein the at least one bracket provides first and second support surfaces adjacent the outer vertical member for receiving the first and second bottom edges of the first and second outer panes, respectively.

**20**. The fire pit surround of claim **19**, wherein at least one bracket defines a first oval orifice receiving a first fastener passing through the first inner pane and the first outer pane, and defines a second oval orifice receiving second fastener passing through the second inner pane and the second outer pane.

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